

Physics 829: Homework Set No. 4

**Due date: Wednesday, April 25, 2012, 1:00pm
in PRB M2043 (Biao Huang's office)**

Total point value of set: 40 points

Problem 1 (20 pts.):

Consider the time-independent Schroedinger equation in 1 dimension for the linear potential $V(x) = g|x|$. We discussed the exact solutions in class, they are Airy functions.

(1) Use the WKB approximation to find an expression for the energy eigenvalue $E_{\text{WKB},n}$ of the n th eigenstate ($n = 0, 1, 2, \dots$) in this potential.

(2) Obtain a variational estimate for the ground state energy in this potential using a Gaussian trial function.

(3) Compare the ground state energies that you obtained from the WKB and variational approximations with the exact value $E_0 = 0.8086 (g^2 \hbar^2 / m)^{1/3}$ and comment. Using MATHEMATICA, plot the normalized exact and optimized variational ground state wave functions and compute their mean square deviation $\int_{-\infty}^{\infty} |\psi_0^{\text{exact}}(x) - \psi_0^{\text{Gauss}}(x)|^2 dx$.

Problem 2 (20 pts.):

The forced harmonic oscillator is defined by the Hamiltonian

$$\hat{H} = \frac{\hat{p}^2}{2m} + \frac{1}{2}m\omega^2 \hat{x}^2 - Q(t)\hat{x} - P(t)\hat{p} = \hbar\omega \left(\hat{a}^\dagger \hat{a} + \frac{1}{2} \right) + f(t)\hat{a} + f^*(t)\hat{a}^\dagger,$$

where

$$f(t) = \int_{-\infty}^{\infty} \frac{d\omega'}{2\pi} f(\omega') e^{i\omega' t} \equiv -\sqrt{\frac{\hbar}{2m\omega}} Q(t) + i\sqrt{\frac{\hbar m\omega}{2}} P(t).$$

For this Hamiltonian, the exact expression for the transition probability is given by the Poisson distribution (see Merzbacher, Sect. 6.6)

$$P_{0 \rightarrow n}(\infty) = |\langle n^0 | \hat{U}_I(\infty, -\infty) | 0^0 \rangle|^2 = \frac{e^{-\langle n \rangle} \langle n \rangle^n}{n!}$$

with mean excitation number $\langle n \rangle = |f(\omega)/\hbar|^2$.

Compute this transition probability in first-order time-dependent perturbation theory and compare with the exact result. Compute also the total energy transferred to the oscillator by the interaction with the external force, both exactly and in first-order perturbation theory. Explain the agreement.

Problem 1

$$\hat{H} = \frac{\hat{p}^2}{2m} + g|x|$$

Schrödinger eqn: $-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} \psi + g|x| \psi = E \psi$

Solutions are parity eigenstates since potential is even.

Odd solutions: $\psi(x=0) = 0$

Even solutions: $\frac{d\psi}{dx}(x=0) = 0$

Solve the eqn. for $x > 0$, connect to $x < 0$ with the above boundary condition at $x=0$.

$x > 0$: substitute $z = \left(\frac{2mg}{\hbar^2}\right)^{1/3} \left(x - \frac{E}{g}\right) \Rightarrow \frac{d^2\psi}{dz^2} - z\psi = 0$
(as shown in class)

normalizability $\Rightarrow \psi(z) \xrightarrow{z \rightarrow \infty} 0$

Solution: Airy function $Ai(z) \xrightarrow{z \rightarrow \infty} \sim \frac{e^{-\frac{2}{3}|z|^{3/2}}}{z^{1/4}}$
(not $Bi(z) \xrightarrow{z \rightarrow \infty} \sim \frac{e^{+\frac{2}{3}|z|^{3/2}}}{z^{1/4}}$)

$\Rightarrow \psi(x > 0) = N Ai\left(\left(\frac{2mg}{\hbar^2}\right)^{1/3} \left(x - \frac{E}{g}\right)\right)$
normalization

The boundary conditions at $x=0$ fix the energy eigenvalues:

even: $\psi'_n(0) = Ai'\left(-\left(\frac{2mg}{\hbar^2}\right)^{1/3} \frac{E_n}{g}\right) = 0 \Rightarrow E_n^{\text{even}}$

odd: $\psi_n(0) = Ai\left(-\left(\frac{2mg}{\hbar^2}\right)^{1/3} \frac{E_n}{g}\right) = 0 \Rightarrow E_n^{\text{odd}}$

To find these eigenvalues, we must look up the zeros of Ai and Ai' .

One finds (Kreuzbacher, p. 124)

n	0	1	2	3	4
$E_n \left(\frac{m}{g^2 \hbar^2}\right)^{1/3}$	0.8086	1.8558	2.5781	3.2446	3.8257

(1) In WKB approximation we find the energy eigenvalues from the condition that the WKB phase integrated between the classical turning points must satisfy the quantization condition

$$\int_a^b k(x) dx = (n + \frac{1}{2})\pi \quad k(x) = \frac{1}{\hbar} \sqrt{2m(E - V(x))}$$

there the classical turning points are $\pm \frac{E_n}{g}$.

$$\Rightarrow \frac{1}{\hbar} \int_{-E_n/g}^{E_n/g} dx \sqrt{2m(E_n - g|x|)} dx = (n + \frac{1}{2})\pi$$

$$\int_0^{E_n/g} dx \sqrt{2m(E_n - gx)} = \frac{\pi \hbar}{2} (n + \frac{1}{2})$$

$$\frac{gx}{E_n} = \zeta$$

$$\int_0^1 \frac{E_n}{g} d\zeta \sqrt{2mE_n} \sqrt{1-\zeta} = \pi (n + \frac{1}{2}) \frac{\hbar}{2}$$

$$\Rightarrow \underbrace{\int_0^1 d\zeta \sqrt{1-\zeta}}_{2/3} = \frac{\pi g (n + \frac{1}{2}) \hbar}{E_n^{3/2} (2m)^{1/2} 2}$$

$$\Rightarrow E_n^{3/2} = \frac{3\pi g \hbar (n + \frac{1}{2})}{4 \sqrt{2m}}$$

$$\Rightarrow E_n^{\text{WKB}} = \left(\frac{g}{32} \pi^2 (n + \frac{1}{2})^2 \right)^{1/3} \left(\frac{g^2 \hbar^2}{m} \right)^{1/3}$$

n	0	1	2	3	4
$E_n^{\text{WKB}} \left(\frac{m}{g^2 \hbar^2} \right)^{1/3}$	0.8853	1.8416	2.5888	3.2397	3.8306

$$*) \int_0^1 d\zeta \sqrt{1-\zeta} \stackrel{\zeta = \sin^2 t}{=} 2 \int_0^{\pi/2} \cos^2 t \sin t dt \stackrel{\cos t = z}{=} 2 \int_0^1 z^2 dz = \frac{2}{3}$$

(2) Variational ground state energy estimate:

Ansatz: $\psi_0(x; \alpha) = \left(\frac{\pi}{\alpha}\right)^{1/4} e^{-\alpha x^2/2} \quad \left(\int_{-\infty}^{\infty} |\psi_0|^2 dx = 1\right)$

$$\langle E \rangle_\alpha = \int_{-\infty}^{\infty} dx e^{-\alpha x^2/2} \left(-\frac{\hbar^2}{2m} \frac{d^2}{dx^2} + g|x| \right) e^{-\alpha x^2/2}$$

$$\frac{d}{dx} (e^{-\alpha x^2/2}) = -\alpha x e^{-\alpha x^2/2}$$

$$\frac{d^2}{dx^2} (e^{-\alpha x^2/2}) = (-\alpha + \alpha^2 x^2) e^{-\alpha x^2/2}$$

$$= +\frac{\alpha^{3/2}}{\sqrt{\pi}} \frac{\hbar^2}{2m} \int_{-\infty}^{\infty} dx (1 - \alpha x^2) e^{-\alpha x^2} + g \sqrt{\frac{\alpha}{\pi}} \int_{-\infty}^{\infty} (2x dx) e^{-\alpha x^2}$$

$$\begin{aligned} & \left(\xi^2 = \alpha x^2 \right) \\ & \left(t = \alpha x^2 \right) \end{aligned} = \frac{\alpha}{\sqrt{\pi}} \frac{\hbar^2}{2m} \underbrace{\int_{-\infty}^{\infty} d\xi (1 - \xi^2) e^{-\xi^2}}_{\sqrt{\pi} - \frac{1}{2}\sqrt{\pi}} + \underbrace{\frac{g}{\sqrt{\pi}\alpha} \int_0^\infty dt e^{-t}}_1$$

$$= \frac{\hbar^2}{4m} \alpha + \frac{g}{\sqrt{\pi}\alpha}$$

This is minimal at $\frac{\hbar^2}{4m} - \frac{1}{2} \frac{g}{\sqrt{\pi}} \alpha^{-3/2} = 0$

$$\Rightarrow \alpha^{3/2} = \frac{4m}{\hbar^2} \frac{g}{2\sqrt{\pi}} = \frac{2mg}{\hbar^2 \sqrt{\pi}} \Rightarrow \boxed{\alpha_0 = \left(\frac{2mg}{\sqrt{\pi} \hbar^2} \right)^{2/3}}$$

$$\Rightarrow \langle E \rangle_0 = \left(\frac{4m^2 g^2 \hbar^6}{\pi \hbar^4 2^6 m^3} \right)^{1/3} + \left(\frac{\sqrt{\pi} \hbar^2 g^3}{2mg \pi^{3/2}} \right)$$

$$= \left(\frac{g^2 \hbar^2}{m} \right)^{1/3} \frac{1}{\pi^{1/3}} \left(\frac{1}{2^{4/3}} + \frac{1}{2^{1/3}} \right) = \left(\frac{g^2 \hbar^2}{m} \right)^{1/3} \frac{1}{(2\pi)^{1/3}} \frac{3}{2} \approx 0.8129 \left(\frac{g^2 \hbar^2}{m} \right)^{1/3}$$

$$\Rightarrow \boxed{E_0^{\text{var}} = 0.8129 \left(\frac{g^2 \hbar^2}{m} \right)^{1/3}}$$

$$(3) \quad E_0^{\text{exact}} = 0.8086 \left(\frac{g^2 \hbar^2}{m} \right)^{1/3}$$

$$E_0^{\text{WKB}} = 0.8853 \left(\frac{g^2 \hbar^2}{m} \right)^{1/3}$$

$$E_0^{\text{var}} = 0.8129 \left(\frac{g^2 \hbar^2}{m} \right)^{1/3}$$

Clearly, for the ground state the variational approach with a Gaussian trial wavefunction works much better than the WKB approximation:

$$\frac{E_0^{\text{WKB}}}{E_0^{\text{exact}}} - 1 = 9.5\%$$

$$, \quad \frac{E_0^{\text{var}}}{E_0^{\text{exact}}} - 1 = 0.5\%$$

The reason is that the Gaussian and exact wave functions are very similar, with

$$\int_{-\infty}^{\infty} |\psi_{\text{exact}}(x) - \psi_{\text{gauss}}(x)|^2 dx \approx 10^{-3}.$$

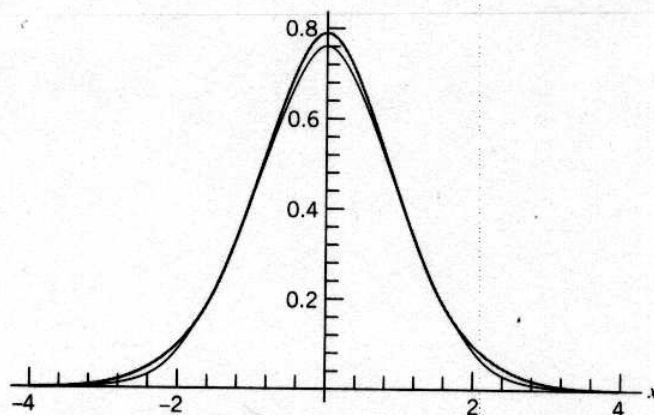


Figure 8.1. The thick line is the normalized ground state energy eigenfunction, $\psi_0(x) = 1.47 \text{ Ai}[\sqrt[3]{2}(x - 0.8086)]$, for the potential $V(x) = |x|$, and the thin line represents the optimal normalized Gaussian variational trial function, $\psi_t(x)$. The x coordinate and the energy are made dimensionless by scaling, as described in Section 7.3. The mean-square deviation between the two functions is only $\int_{-\infty}^{+\infty} |\psi_0(x) - \psi_t(x)|^2 dx \approx 10^{-3}$!

Problem 2

$$\hat{H} = \hbar\omega(\hat{a}^\dagger\hat{a} + \frac{1}{2}) + \underbrace{f(t)\hat{a} + f^*(t)\hat{a}^\dagger}_{\hat{H}_1(t)} = \hat{H}_0 + \hat{H}_1(t)$$

We are given the exact result

$$P_{0 \rightarrow n}(\infty) = |\langle n^0 | \hat{U}_I(\infty, -\infty) | 0^0 \rangle|^2 = \frac{e^{-\langle n \rangle} \langle n \rangle^n}{n!}$$

$$\text{with } \langle n \rangle = \left| \frac{f(\omega)}{\hbar} \right|^2, \quad f(\omega) = \int_{-\infty}^{\infty} dt e^{-i\omega t} f(t)$$

In first order perturbation theory we get

$$\begin{aligned} c_n^{(1)}(\infty) &= \delta_{n0} - \frac{i}{\hbar} \int_{-\infty}^{\infty} \langle n^0 | f(t)\hat{a} + f^*(t)\hat{a}^\dagger | 0^0 \rangle e^{i\omega t} dt \\ &= \delta_{n0} - \frac{i}{\hbar} \delta_{n1} \underbrace{\langle 1^0 | \hat{a}^\dagger | 0^0 \rangle}_1 \int_{-\infty}^{\infty} dt e^{i\omega t} f^*(t) \\ &= \delta_{n0} - \frac{i}{\hbar} \delta_{n1} f^*(\omega) \end{aligned}$$

$$\Rightarrow \boxed{P_{0 \rightarrow 0}^{(1)}(\infty) = 1, \quad P_{0 \rightarrow 1}^{(1)}(\infty) = \left| \frac{f(\omega)}{\hbar} \right|^2}$$

If we expand the exact result to lowest nontrivial order in $f(\omega)$, we get

$$P_{0 \rightarrow 0}(\infty) = e^{-|f(\omega)/\hbar|^2} = 1 - \left| \frac{f(\omega)}{\hbar} \right|^2 + \dots \approx 1$$

$$P_{0 \rightarrow 1}(\infty) = e^{-|f(\omega)/\hbar|^2} \left| \frac{f(\omega)}{\hbar} \right|^2 \approx \left| \frac{f(\omega)}{\hbar} \right|^2$$

$$P_{0 \rightarrow (n>1)}(\infty) \sim \left| \frac{f(\omega)}{\hbar} \right|^{2n} \approx 0$$

in agreement with the perturbative result, within the accuracy of the approximation.

The energy transferred to the oscillator is in pert. theory

$$E_{\text{abs}}^{(1)} = \frac{\sum_n P_{0 \rightarrow n}^{(1)}(\infty) (n \hbar \omega)}{\sum_n P_{0 \rightarrow n}^{(1)}} = \frac{0 + \hbar \omega \left| \frac{f(\omega)}{\hbar} \right|^2}{1 + 0 \left(\left| \frac{f(\omega)}{\hbar} \right|^2 \right)} = \hbar \omega \left| \frac{f(\omega)}{\hbar} \right|^2$$

The exact result gives

$$\begin{aligned} E_{\text{abs}} &= \frac{\sum_n P_{0 \rightarrow n}(\infty) (n \hbar \omega)}{\sum_n P_{0 \rightarrow n}} = \hbar \omega \sum_n n P_{0 \rightarrow n} \\ &= \langle n \rangle \hbar \omega = \left| \frac{f(\omega)}{\hbar} \right|^2 \hbar \omega \end{aligned}$$

This agrees exactly with the perturbative result!

This is surprising, since the probability to excite higher-lying states is non-zero beyond leading order in perturbation theory. So as we increase the accuracy of our approximation, the amount of energy transferred by exciting the first excited state is reduced (by reducing the excitation probability) in just such a way that this compensates for energy transferred by exciting higher-lying states.

Since the energy eigenvalues of the harmonic oscillator are equidistant, the mean transferred energy is two times the mean excitation level $\langle n \rangle$. For both the exact Poisson distribution and the distribution of excitation levels in 1st order perturbation theory, $\langle n \rangle = \left| \frac{f(\omega)}{\hbar} \right|^2$. This explains why both give the same transferred energy.